Integral Applications: Study Guide

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1 Concrete Lebesgue and Hausdorff integral applications

A first collection of **16 fully worked examples**, written for an analysis student. Start with the problem index, then open the separate solution that you want. The examples favor explicit computations, stated hypotheses, and direct arguments explaining why the integration theorem applies.

- Measures and theorem conventions: normalized Hausdorff measure, completed-product sections, signed integrability, area and coarea.
- Problem index: no computed answers.
- Counterexample atlas: precise failures and sufficient conditions for exchanging sums, limits, derivatives, and integrals.
- Full worked collection, PDF and editable LaTeX.
- Compact guide, PDF and LaTeX.
- Source references, source snapshot hashes, and verification ledger.
- Machine-readable index without results, and full metadata with hypotheses, results, and review status.

1.1 Find a technique again

From the repository root, using Python 3 with its standard library:

```
python3 integral-examples/find_examples.py --topic coarea
python3 integral-examples/find_examples.py --technique multiplicity
python3 integral-examples/find_examples.py --prerequisite dominated
python3 integral-examples/find_examples.py --text concentration
python3 integral-examples/find_examples.py --list-tags
```

Filters are case-insensitive substring matches and combine with AND. They print titles, tags, and paths relative to `integral-examples/`. The `--text` filter searches IDs, titles, paths, topics, prerequisites, and techniques; it does not search solution prose or recorded hypotheses/results. Add `--solutions` to print the recorded result. The files remain ordinary Markdown and JSON; retrieval needs no hosted service.

1.2 How to study with the collection

For each problem, identify the function, its domain, the measure, and the integrability question before computing. In a region problem, write the indicator and determine its sections. For signed integrals, examine the absolute value or the positive and negative parts. For a parametrization, compute the Jacobian and count preimages before identifying the answer with an integral over the image. For a parameter limit, write the proposed majorant explicitly.

E01–E06 cover Lebesgue integration and iterated integrals; E07–E13 cover curves, surfaces, area, coarea, and singular maps; E14–E16 cover limit and derivative exchanges and approximate identities. Cross-links in the atlas make failure cases easy to retrieve alongside their positive criteria.

1.3 Reproduce the checks and PDFs

```
python3 integral-examples/checks/check_examples.py
python3 integral-examples/checks/check_collection.py
python3 integral-examples/build.py
```

The concrete identity checker uses SymPy and mpmath. The collection checker and retrieval command use only the Python standard library. The build uses Pandoc and pdfLaTeX. See checks/requirements.txt for the versions actually used. The checked-in PDF and LaTeX need no build tools to read.

Independent model reviewers audited every example; the ledger records the scope, findings, and

revisions. Symbolic and numerical checks supplement those reviews and are **not proofs of analytic hypotheses**. No Lean verification or human peer review is claimed. General measure-theoretic theorems are cited prerequisites; this collection does not reprove all of measure theory.

All collection changes are confined to `integral-examples/` on branch `codex/integral-examples-2026-10-15`, based on the integrated research commit `e4f337c3aa52ec4c3f2aab078d7c91b32ee0015f`. Other research branches are independent. The earlier paper was not used or reconstructed. No third-party problem collection or source PDF is reproduced, and no license is assigned here to the cited sources or to unrelated repository work.

2 Problem index — without computed answers

Each entry links to a separate full solution. Read CONVENTIONS.md first for measure normalization and the exact role of almost-everywhere sections. The JSON index and retrieval command described in README.md support search by topic, prerequisite, and technique without printing answers by default.

ID	Task	Main prerequisites	Technique to look for
E01	Integrate e^{y^2} over $0 < x < y < 1$.	Measurability, Tonelli	Change order on a measurable region
E02	Construct a region whose boundary has positive area; integrate $2y$ over it.	Countable additivity, Tonelli	Explicit removed-interval sums
E03	Can every section of a Lebesgue-measurable null set be integrated? Prove the correct general statement.	Completion, Fubini	Borel representative and null envelope
E04	Compare the iterated integrals of $(x^2 - y^2)/(x^2 + y^2)^2$ on $(0, 1)^2$.	Signed integrals, Fubini	Check positive and negative parts first
E05	Sum an explicit signed matrix by rows and by columns.	Counting measure, series	Detect infinite absolute mass
E06	Evaluate and differentiate a Frullani parameter integral.	Tonelli, dominated convergence	Replace a quotient by a parameter integral
E07	Integrate the first coordinate along a parabola with respect to length.	Derivatives, area formula	Compute the speed and pull back the weight
E08	Integrate a coordinate square on a twice-traversed circle.	Curve area formula	Count multiplicity; distinguish circulation
E09	Integrate height on a paraboloid graph over the unit disk.	Gram determinant, polar coordinates	Derive the graph area factor
E10	Integrate a coordinate square over a sphere of radius R.	Surface area formula	Handle chart seams and pole exceptions
E11	Apply weighted area to a folded square parametrization.	Jacobian, area formula	Compare source integral and image multiplicity
E12	Turn an integral on a disk into integrals over circular level sets.	Gradient, coarea	Account for a critical point before division

ID	Task	Main prerequisites	Technique to look for
E13	Explain why the Cantor staircase violates a proposed derivative-area identity.	Cantor set, absolute continuity	Locate the failed Lipschitz hypothesis
E14	Track mass in concentration and escape examples when taking limits under integrals.	Pointwise convergence, DCT	Uniform integrability and spatial tightness
E15	Compare a parameter derivative with the integral of pointwise derivatives.	Difference quotients, DCT	Test local domination; distinguish derivative limits
E16	Recover a boundary value from exponential averages.	Substitution, DCT	Approximate identity and an explicit tail estimate

Suggested routes: E01–E06 for Lebesgue integration and iterated integrals; E07–E13 for geometric integration; E14–E16 for exchanges of limits, derivatives, and integrals. E03 and E04 are useful safeguards before generalizing a successful calculation. The counterexample atlas maps the safeguards to precise sufficient conditions and contains some outcome spoilers.

3 Measures, functions, and theorems used in this library

The examples are applications for an analysis student. Their proofs are written out; the general integration theorems below are prerequisites, with sources in SOURCES.md. A computation check verifies an identity, not the measurability or integrability hypotheses that justify it.

3.1 Functions and values

A function is a map with a specified domain, codomain, and rule. For example, $f : (0,1)^2 \to \mathbb{R}$ is a function, $f(x,y)$ is its value at (x,y), and $f_x : y \mapsto f(x,y)$ is a section function. With a fixed integrand and measure, the integral is a number (or an allowed extended value), not another notation for the function. A parameter integral defines a new function, for example $F : t \mapsto \int f_t \, d\mu$; $F(t)$ is its value. When a section integral is defined only almost everywhere, a measurable version is chosen before integrating that version in the outer variable.

3.2 Lebesgue and counting measures

$\mathcal{L}^n$ denotes complete n-dimensional Lebesgue measure on $\mathbb{R}^n$. A restricted integral $\int_E f \, d\mathcal{L}^n$ means integration over the measurable set E with the restricted measure. For finite-valued f defined on all of $\mathbb{R}^n$, it equals $\int f \mathbf{1}_E \, d\mathcal{L}^n$. Counting measure on $\mathbb{N} = \{1, 2, \ldots\}$ turns integration into an explicit sum. The Lebesgue and countable counting measure factors used in the product theorems here are sigma-finite. This does not assert sigma-finiteness of $\mathcal{H}^k$ on all of $\mathbb{R}^n$ when $k < n$; our curve and surface examples use finite-measure or countably covered restrictions.

For real measurable f, set $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. A finite Lebesgue integral requires

$$\int |f| \, d\mu = \int f^+ \, d\mu + \int f^- \, d\mu < \infty.$$

An extended signed integral is defined if at least one of the two terms is finite. If both are infinite, their difference is undefined. An improper limit or principal value must be named separately; its existence does not supply a signed Lebesgue integral.

3.3 Normalized Hausdorff measure

For an integer $k \geq 1$, let $\omega_k = \mathcal{L}^k(B(0,1))$. We use

$$\mathcal{H}^k(E) = \lim_{\delta \downarrow 0} \inf \left\{ \sum_{i=1}^{\infty} \omega_k \left(\frac{\operatorname{diam} U_i}{2} \right)^k : E \subseteq \bigcup_{i=1}^{\infty} U_i, \quad \operatorname{diam} U_i \leq \delta \right\}.$$

$\mathcal{H}^0$ is counting measure. This normalization makes $\mathcal{H}^k$ agree with $\mathcal{L}^k$ on every k-plane, using an isometry to identify that plane with $\mathbb{R}^k$. Thus $\mathcal{H}^1$ measures length and $\mathcal{H}^2$ measures surface area. In particular $\omega_1 = 2$ and $\omega_2 = \pi$. Some books omit the factor $\omega_k/2^k$; their numerical values must be rescaled before using these examples. All Hausdorff integrals in this library use the convention in this paragraph.

3.4 Tonelli and Fubini, including completion

For a nonnegative product-measurable function on sigma-finite measure spaces, Tonelli permits either order of integration, including the value $+\infty$. For a signed or complex product-measurable function, Fubini applies if the absolute value has finite integral. It then gives finite section integrals almost everywhere and equality of the two iterated integrals with the product integral.

The product of two complete Lebesgue sigma-algebras need not be complete. For $\mathcal{L}^{p+q}$-measurable sets or functions on $\mathbb{R}^{p+q}$, use the completed-product versions: sections need be measurable only for almost every parameter. A Borel set has Borel sections at every parameter; an arbitrary Lebesgue-measurable set need not have measurable sections at every parameter. The outer integral uses a measurable representative of the almost-everywhere section-integral function. E03 proves this distinction directly with a null envelope. No claim about all sections is inferred from completeness.

3.5 Area with multiplicity

For a Lipschitz map $F : \mathbb{R}^m \to \mathbb{R}^n$ with $m \leq n$, write

$$J_m F(x) = \sqrt{\det(DF(x)^T DF(x))}, \qquad N(F, A, z) = \#\{x \in A : F(x) = z\}.$$

The derivative and Jacobian exist almost everywhere; set the Jacobian to zero at the exceptional points where the derivative does not exist. For a measurable domain A and nonnegative Borel weight h on the source, the weighted area formula is

$$\int_A h(x) J_m F(x)\, d\mathcal{L}^m(x) = \int_{\mathbb{R}^n} \left(\sum_{x \in A \cap F^{-1}(\{z\})} h(x) \right) d\mathcal{H}^m(z).$$

The sum is an extended nonnegative sum. With $h = 1$ it counts multiplicity. With a nonnegative Borel w and $h = w \circ F$ it becomes $w(z)N(F, A, z)$, with the nonnegative-integration convention $0 \cdot \infty = 0$. An injective parametrization therefore computes the geometric scalar integral once. If

multiplicity is q almost everywhere, the parameter integral counts it q times. For signed weights require finite absolute integral with this multiplicity. A map on a bounded domain that is Lipschitz there can be extended to apply the formula; the geometric parametrizations to which we apply area are also Lipschitz on a suitable containing compact box. The Cantor staircase is an explicitly excluded counterexample.

For a graph $F(u) = (u, g(u))$ with a scalar C^1 function g, $J_m F = \sqrt{1 + |\nabla g|^2}$. This factor is derived in the graph example.

3.6 Coarea and division by a Jacobian

For a Lipschitz map $G : \mathbb{R}^n \to \mathbb{R}^m$, $n \geq m$, the coarea Jacobian is

$$J_m G(x) = \sqrt{\det(DG(x)DG(x)^T)}.$$

For nonnegative Borel h, coarea says

$$\int_{\mathbb{R}^n} h(x) J_m G(x)\, d\mathcal{L}^n(x) = \int_{\mathbb{R}^m} \left(\int_{G^{-1}(\{t\})} h(x)\, d\mathcal{H}^{n-m}(x) \right) d\mathcal{L}^m(t).$$

Fiber statements and inner integrals have their usual almost-everywhere meaning. Here $m = 1$ gives $J_1 G = |\nabla G|$. To recover an unweighted volume integral, division by $J_m G$ is legitimate only on its positive set. Any contribution from its zero set must be handled separately. E12 identifies and disposes of its critical point explicitly. E13 shows why a singular continuous map cannot be substituted into a Lipschitz area formula.

3.7 Scalar integration and orientation

$\int_S w\, d\mathcal{H}^k$ has no orientation. A scalar curve parametrization uses speed $|\gamma'(t)|$. A vector circulation uses $\int F(\gamma(t)) \cdot \gamma'(t)\, dt$; reversing traversal changes its sign. Flux through an oriented surface is $\int_S F \cdot \nu\, d\mathcal{H}^{n-1}$ with a chosen unit normal ν. Changing ν changes flux. Neither a tangent vector nor a normal vector is part of Hausdorff measure itself.

3.8 Convergence and differentiation

The version of monotone convergence used here concerns nonnegative increasing measurable functions. Dominated convergence concerns measurable functions and requires almost-everywhere convergence and one integrable majorant for the entire sequence. Pointwise convergence, boundedness of each individual function, or a uniform bound on the integrals alone is insufficient. E14 gives concentration and escape examples with explicit sufficient repairs. E15 states and proves a local domination criterion for differentiating an integral. E16 explains how concentration can instead be used deliberately as an approximate identity.

4 A small atlas of failed exchanges

This is a retrieval map to complete proofs, not a replacement for them. It gives outcomes; use PROBLEMS.md for the problem-first route.

Exchange or tempting inference	Example and failure mechanism	Positive sufficient condition
Integrate every section of a Lebesgue-measurable set	E03: a nonmeasurable set can sit in one exceptional section of a planar null set.	Borel measurability gives Borel sections everywhere; completed Lebesgue measurability gives measurable sections almost everywhere.
Swap signed iterated integrals	E04: both signed parts have infinite integral near the origin. E05: the signed matrix has infinite absolute sum.	Joint measurability and finite integral of the absolute value give Fubini; nonnegativity gives Tonelli with extended values.
Pass an almost-everywhere limit through an integral	E14: mass concentrates at a shrinking interval; its height grows.	DCT with one integrable majorant; on a finite measure space, convergence in measure plus uniform integrability gives convergence in L^1.
Treat bounded integral mass as sufficient on an infinite space	E14: a unit bump moves to infinity. Value-tail uniform integrability alone permits this.	Add spatial tightness to value-tail uniform integrability, together with convergence in measure on bounded sets (and an integrable limit). Precise versions are stated in E14.
Differentiate a parameter integral at an endpoint	E15: difference-quotient mass concentrates near zero; pointwise derivatives miss that mass.	Measurability, pointwise differentiability almost everywhere, one integrable local bound on the parameter derivative, and one integrable parameter slice. See the exact interval and endpoint conditions in E15.
Differentiate a uniform limit of differentiable functions	E15 related example: $\sin(nx)/n$ tends uniformly to zero while its derivative is 1 at zero for every n.	On a compact interval, uniform convergence of the derivatives and convergence of the function values at one point imply a differentiable limit with the expected derivative. The mean value theorem proves this for differentiable functions; for C^1 functions one can also use the fundamental theorem of calculus.
Replace geometric area by parameter area	E08/E11: repeated or folded parametrizations count the image more than once.	Injectivity outside sets whose contributions are zero, or explicit multiplicity in the area formula.

Exchange or tempting inference	Example and failure mechanism	Positive sufficient condition
Divide by a coarea Jacobian everywhere	E12: the gradient vanishes at the origin.	Restrict division to the positive-Jacobian set, then prove the zero set contributes zero or account for its contribution separately.
Use the area formula for a merely continuous monotone map	E13: the Cantor staircase has derivative zero almost everywhere but maps onto an interval.	Lipschitz maps satisfy the area formula; in one dimension absolute continuity gives the fundamental theorem and the relevant change-of-variable identities.

Concentration is also a tool: E16 uses an approximate identity. There the integration variable is rescaled and a fixed integrable majorant becomes available. The conclusion is a boundary evaluation, with all its hypotheses stated, rather than an exchange based solely on pointwise convergence in the original variable.

5 Sources and conventions audit

All links below were opened and checked on **2026-10-15**. These are author-hosted teaching texts, used for theorem statements and conventions. The worked examples in this directory are original expositions; the linked texts' proofs and problem collections are not reproduced. PDF page numbers below count the cover as page 1. A PDF viewer may display a different logical page label. Online files can change.

5.1 `hunter-measure`

John K. Hunter, *Measure Theory*, University of California, Davis, 2011. Author's PDF.

- Monotone convergence: Theorem 4.6, printed p.41, PDF p.45.
- Dominated convergence: Theorem 4.24, printed p.49, PDF p.53; a.e. versions are discussed in §4.6, printed p.50.
- Product-measurable sections: §5.1, Proposition 5.2, printed p.55, PDF p.59; function sections: Theorem 5.15, printed p.60.
- Fubini with absolute integrability: Theorem 5.18, printed p.61, PDF p.65.
- Completion and exceptional sections: Example 5.20 and Theorem 5.21, printed p.62, PDF p.66.

The completed Lebesgue product statement guarantees measurable and integrable sections almost everywhere. It does not guarantee measurability of every section. Theorem 5.15's sentence defining complex section integrals suppresses existence hypotheses; use the explicit integrability hypotheses of 5.18/5.21.

5.2 `axler-mira`

Sheldon Axler, *Measure, Integration & Real Analysis*, Graduate Texts in Mathematics 282, Springer, 2020; author's online revision dated 12 June 2026. Author's PDF.

- Tonelli: §5B, Theorem 5.28, printed p.129, PDF p.144.
- Nonnegative double sums: Corollary 5.31, printed p.131, PDF p.146.
- Fubini: Theorem 5.32, printed p.132, PDF p.147, with exceptional-parameter conventions immediately above and in the proof.

These product theorems assume sigma-finite factors. Tonelli accepts nonnegative product-measurable functions and permits the value infinity; Fubini requires a finite integral of the absolute value. This is an independent reference for the distinction used in the examples. For completed Lebesgue measurability rather than product measurability, also use `hunter-measure` §5.7.

5.3 `simon-gmt`

Leon Simon, *Introduction to Geometric Measure Theory*, Stanford University, 2017/2018 NTU lecture draft; last-update line: 7 March 2018. Author's PDF.

- Normalization: Chapter 1, definitions 2.1–2.2, printed p.11, PDF p.9; equality with Lebesgue outer measure: Theorem 2.6, printed p.12, PDF p.10.
- Area and multiplicity: Chapter 2, Theorem 3.3, printed p.57, PDF p.32; weighted fiber sum: Remark 3.4(2), printed p.58, PDF p.33; curves and submanifolds: Examples 3.5, printed pp.60–61, PDF pp.34–35.
- Coarea: Chapter 2, Theorem 7.3 and weighted Remark 7.4(1), printed p.76, PDF p.42.
- Critical parts of fibers: Theorem 7.5, printed p.80, PDF p.44; the stronger smooth Sard conclusion is distinguished in Remark 7.6.

The PDF has two printed pages per sheet. This version uses the factor `omega_k / 2^k` multiplying sums of diameter powers, where `omega_k` is unit-ball volume. Consequently normalized Hausdorff measure agrees with Lebesgue measure on k-planes. Counting measure is the k=0 convention.

5.4 `hajlasz-geometry`

Piotr Hajłasz, *Geometric Analysis*, University of Pittsburgh, undated author-hosted lecture notes. Author's PDF.

- Normalization: §3.1, printed/PDF p.19; equality with Lebesgue measure: Theorem 3.2, p.20.
- Surface integration: Theorem 3.17 and following discussion, p.25.
- Area and multiplicity with the Lusin N condition: Theorem 3.18 and Remark 3.19, pp.26–27.
- Coarea: §3.4, Theorem 3.23, pp.28–29; radial slicing and projection/Fubini applications, p.29. Almost-everywhere rectifiability: Theorem 3.24, p.29.

Normalization agrees with `simon-gmt`. Two evident dimensional misprints occur in §3.4: the coarea Jacobian should vanish when the derivative has rank less than **m**, not n; the area half of Theorem 3.23 needs outer measure **H^n**, not H^m. The coarea half uses the correct fiber and target dimensions. Our area statements use Simon's 2018 Theorem 3.3 and Remark 3.4(2), whose dimensions are correct.

5.5 `hajlasz-analysis`

Piotr Hajłasz, *Analysis I*, University of Pittsburgh, undated author-hosted lecture notes. Author's PDF.

- Theorem 103 and the Cantor staircase discussion, printed/PDF p.76: a continuous increasing function can have derivative zero almost everywhere while its endpoint increment is positive.
- Absolute continuity and the fundamental theorem: §8.1, Theorem 106, printed/PDF pp.77–78.
- Convolution in L1: Theorem 87, pp.66–67; smooth approximate identities: Theorem 92, pp.70–71.

The Cantor paragraph prints `f'(0) = 0 a.e.`; the variable should be x. The example in this library proves its needed staircase properties directly. Derivative values on its exceptional null set may be assigned arbitrarily for Lebesgue integration, but those assignments do not make the staircase absolutely continuous or Lipschitz.

5.6 `conrad-diff`

Keith Conrad, *Differentiating Under the Integral Sign*, University of Connecticut, undated author-hosted expository notes. Author's PDF.

- A sufficient differentiation theorem: §12, Theorem 12.3, printed/PDF p.16.
- Local parameter bounds illustrating the theorem: Table 2, p.16, and discussion on p.17.

Theorem 12.3 treats continuous integrands and parameter derivatives, dominated by parameter-independent integrable majorants near the parameter in question. Its infinite-interval integrals are stated as improper integrals. The library supplies Lebesgue proofs via dominated convergence of difference quotients; it does not silently equate conditional improper integration with Lebesgue integrability.

5.7 Reading formulas across versions

Use the source IDs above, not a theorem number alone: Simon's 2014 and 2018 drafts number area and coarea differently. The older 2014 course draft, printed p.54, formula 3.4, also has the incorrect target dimension Hˆm; the following weighted formula 3.5 has Hˆn. The library cites the corrected 2018 area statement.

The convention in this directory is scalar integration against positive measures. A parametrization's area Jacobian is nonnegative and counts repeated coverings. An oriented vector flux additionally needs a chosen normal and a signed dot product; it is not determined by the scalar Hausdorff measure alone.

No source access, symbolic calculation, numerical test, or review recorded here is a claim of proof-assistant verification.